

Problem Set 5

Probability and Statistical Inference

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1. Let $Z \sim N(0, 1)$, let $I_n \sim \text{Bernoulli}(1/n^2)$, and let U equal -1 or 1 with equal probability. The following are three sequences of estimators for a fixed target β :

$$A_n = \beta + \frac{Z}{\sqrt{n}}, \quad B_n = \beta + nI_n, \quad C_n = \beta + U.$$

- (a) For each estimator, calculate its bias, variance, and mean squared error.
 - (b) For each sequence, determine whether it converges to β in mean square and whether it converges to β in probability. Also state its limiting distribution.
 - (c) Find the distribution of $\sqrt{n}(A_n - \beta)$ for every n . Explain what this tells us beyond the statement $A_n \xrightarrow{p} \beta$.
 - (d) Which sequence shows that convergence in probability need not imply convergence in mean square? Explain the role of rare, large errors.
 - (e) You may use the following probability bound: $\Pr(W > a) \leq \mathbb{E}[W]/a$ for any random variable $W \geq 0$ and any $a > 0$. For an arbitrary sequence T_n , apply it to $(T_n - \beta)^2$ to prove that $T_n \xrightarrow{\text{m.s.}} \beta$ implies $T_n \xrightarrow{p} \beta$.
2. This question explores why both parts of “independent and identically distributed” matter.
- (a) Suppose $X_i = \mu + A + e_i$. Here A is one mean-zero common shock, with finite variance, that appears in every observation. The e_i are iid with mean zero and finite variance, and A is independent of every e_i . Find the probability limit of $\bar{X}_n$. For $i \neq j$, calculate $\text{Cov}(X_i, X_j)$ and explain why averaging does not remove A .
 - (b) Now suppose the X_i are independent and $X_i \sim N(0, i^a)$, where the second argument is the variance and $a \geq 0$. Show that

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n i^a, \quad \frac{\text{Var}(\bar{X}_n)}{n^{a-1}/(a+1)} \longrightarrow 1.$$

You may compare the sum with $\int_0^n x^a dx$ and $\int_1^{n+1} x^a dx$. For which values of a does $\bar{X}_n \xrightarrow{p} 0$? What happens when $a = 1$ and when $a > 1$?

- (c) A researcher samples only from one friendship network. The sample average approaches $\mu + g$, but the mean in the target population is μ . Explain separately the problem caused by dependence and the problem caused by the shift g . Must a larger sample from the same network solve either problem?
 - (d) State weak stationarity in terms of the mean, variance and covariances at different lags of a time series. Briefly explain why stationarity together with suitable weak dependence can replace iid sampling in a time-series law of large numbers.
3. Consider the linear model $y_i = x_i' \beta + \varepsilon_i$, where x_i is a $k \times 1$ vector and the data matrix X has full column rank (A1 and A2). Assume the expectations used below exist.

- (a) Starting from $\hat{\beta} = (X'X)^{-1}X'y$, derive the exact decomposition of $\hat{\beta} - \beta$. Under A3, $\mathbb{E}(\varepsilon | X) = 0$, establish finite-sample conditional and unconditional unbiasedness.
- (b) Rewrite the decomposition using sample averages. Suppose laws of large numbers give

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q, \quad \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} g,$$

where Q is nonsingular. Find $\text{plim } \hat{\beta}$ and state the condition for consistency. You may use the probability-limit rules for products and inverses used in the lecture's OLS-consistency argument; you do not need to state or prove the named theorems behind these rules.

- (c) Explain why $\mathbb{E}(\varepsilon_i | x_i) = 0$ implies $\mathbb{E}(x_i \varepsilon_i) = 0$. Why is the latter condition weaker?
- (d) Let

$$Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 0.1 \\ 0.4 \end{pmatrix}.$$

Calculate $Q^{-1}g$ and describe where the intercept and slope estimates converge relative to their targets.

- (e) Does the algebraic decomposition change for time-series data? Which assumptions used to justify the probability limits must change?
4. This question uses the central limit theorem for approximation and inference.
- (a) Let X_i be iid exponential random variables with rate 1, so that $\mathbb{E}(X_i) = \text{Var}(X_i) = 1$. For $n = 100$, approximate $\Pr(0.8 < \bar{X}_n < 1.15)$. You may use $\Phi(1.5) = 0.9332$ and $\Phi(-2) = 0.0228$. Why can a normal approximation be useful even though an individual exponential draw is strongly right-skewed?
- (b) In a scalar regression without an intercept, assume iid observations, A1, A2, A3' and the finite-moment conditions for Lecture 5's OLS normality result. Suppose

$$Q = \mathbb{E}(x_i^2) = 2, \quad S = \mathbb{E}(x_i^2 \varepsilon_i^2) = 18.$$

Use the OLS asymptotic variance formula $V = Q^{-1}SQ^{-1}$ to find V . For $n = 200$ and $\hat{\beta} = 1.30$, calculate the asymptotic standard error, a 95% confidence interval, the two-sided statistic for $H_0 : \beta = 1$, and its approximate two-sided p -value. You may use $z_{0.975} = 1.96$ and $\Phi(2) = 0.97725$.

- (c) Explain why S , rather than the homoscedastic simplification $\sigma^2 Q$, is the appropriate object in part (b).

5. Suppose

$$Z_n = \sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} Z,$$

where Z has a distribution on the real line that is not concentrated at a single value. You may use the following probability bound: for every $\eta > 0$, there are finite $M > 0$ and N such that $\Pr(|Z_n| > M) < \eta$ for all $n \geq N$. In words, the scaled error stays within a fixed range with arbitrarily high probability once n is large enough.

- (a) Use the supplied bound and the definition of consistency to prove that $\hat{\beta}_n \xrightarrow{p} \beta$.
- (b) Explain the difference between the two conclusions: consistency concerns the unscaled error, while asymptotic normality (or another distributional limit) concerns the shape of a scaled error.
- (c) For these comparisons, use the working approximation that a typical error or standard error has the form $c/\sqrt{n}$. What happens when the sample size is multiplied by 4? By 9? By what factor must the sample size increase to reduce the error or standard error by 40%?

Non-examinable extension (for self-learning only)

6. Let $\beta > 0$ and suppose $\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(0, V)$, with $V > 0$. Assume the estimates $\hat{\beta}_n$ and $\hat{V}$ used below are positive.
- (a) The delta method states that the asymptotic variance after a differentiable transformation h is $[h'(\beta)]^2 V$. Apply it to $h(\beta) = \log \beta$.
 - (b) If $\hat{V} \xrightarrow{p} V$ and $\hat{\beta}_n \xrightarrow{p} \beta$, give a consistent estimated standard error for $\log \hat{\beta}_n$ and a test statistic obtained by dividing the estimation error by that estimated standard error. Use Slutsky's theorem to state its limit.